EHR Clinical Intelligence Platform

Week 1 Technical Artifacts: Multi-Source Heterogeneous Ingestion, Canonical Data Models & Clinical NLP Baseline

Author: AI Clinical Engineering Team	**Status: Verified & In Production**	**Generated: 2026-09-18**
Target Services: Ingestion (8000), Terminology (8001)	**Infrastructure: PostgreSQL (5432, 5433), MinIO S3 (9000)**	**Test Suite: 6/6 Integration Tests Passing**

1. System Topology & Core Infrastructure

Week 1 deployed the foundational persistence layer separating operational EHR transactions from high-dimensional terminology vector lookups. All components run as containerized microservices managed through Docker Compose meshes.

Component	Technology	Port	Schema / Bucket / Role
Primary Operational DB	PostgreSQL 16	`5432`	Database: ehr_primary (patients, encounters, documents, fragments)
Terminology Vector DB	PostgreSQL 16 + pgvector	`5433`	Database: terminology_db (384-dim dense cosine embeddings)
Object Storage Engine	MinIO S3 API	`9000 / 9001`	Bucket: ehr-raw-inputs (SHA-256 verified raw payloads)
Ingestion Microservice	FastAPI / Python 3.11	`8000`	REST API: Multi-adapter ingestion, seeding, queries
Terminology Microservice	FastAPI / Python 3.11	`8001`	Hybrid lexical & vector semantic search over ontologies

2. Canonical Relational Models (services/ingestion/models.py)

The relational schema decouples heterogeneous input streams into normalized entities with full foreign key constraints and audit indexes.

```
-- Canonical Source Document Catalog
CREATE TABLE source_documents (
    id VARCHAR PRIMARY KEY,               -- e.g. STT-2026-081234
    patient_id VARCHAR REFERENCES patients(id), -- NULL if quarantined
    encounter_id VARCHAR REFERENCES encounters(id),
    source_type VARCHAR NOT NULL,          -- SPEECH_TO_TEXT, OCR, HISTORICAL_DOCUMENT, HL7_API_JSON
    document_class VARCHAR,                -- consultation_transcript, prescription
    minio_path VARCHAR NOT NULL,           -- S3 URI in ehr-raw-inputs
    sha256_checksum VARCHAR(64) NOT NULL,  -- 64-char hexadecimal digest
    raw_text TEXT NOT NULL,
    identity_hint_fields JSONB,            -- Demographic hints for EMPI
    status VARCHAR DEFAULT 'INGESTED'      -- Lifecycle: INGESTED, PENDING_IDENTITY_RESOLUTION, LINKED
);

-- Atomic Clinical Fragments
CREATE TABLE source_fragments (
    id VARCHAR PRIMARY KEY,
    document_id VARCHAR REFERENCES source_documents(id) ON DELETE CASCADE,
    fragment_index INTEGER NOT NULL,
    text_content TEXT NOT NULL,
    confidence FLOAT DEFAULT 1.0,
    speaker_turn VARCHAR,                  -- DOCTOR | PATIENT
    bounding_box JSONB                     -- [ymin, xmin, ymax, xmax]
);
```

3. Heterogeneous Multi-Source Ingestion Adapters

The ingestion engine supports 4 distinct clinical stream types, each handled by an isolated adapter in services/ingestion/adapters/:

Adapter	Source Format	Extraction Logic & Attributes	Quarantine Policy
STTAdapter	Diarized Audio JSON	Extracts speaker turns (DOCTOR/PATIENT), start/end millisecond timestamps, acoustic confidences.	Immediate Link (status = INGESTED)
OCRAdapter	Scanned Prescriptions	Extracts text blocks, OCR vision confidence scores, and bounding box coordinates [ymin, xmin, ymax, xmax].	Current Encounter (status = INGESTED)
HistoricalAdapter	Legacy Hospital PDFs / OCR	Extracts demographic hints (Name, DOB, Gender, MRN, Facility) into identity_hint_fields.	Strict Quarantine (status = PENDING_IDENTITY_RESOLUTION, patient_id = None)
HL7APIAdapter	FHIR / HL7 v2 JSON	Unpacks Condition, Observation, MedicationRequest, and DiagnosticReport resource collections.	Immediate Link (status = INGESTED)

> ***Architectural Safety Invariant:*** *Per EHR Architecture Sec 4.3, external historical records are strictly quarantined on arrival. They are assigned patient_id = None and status = 'PENDING_IDENTITY_RESOLUTION' to guarantee zero leakage into the active patient chart prior to verified identity resolution.*

4. Clinical AI & NLP Baseline Extractor

Implemented in services/clinical-processing/nlp/extractor.py, the engine performs multi-class clinical entity extraction, assertion analysis (NegEx scope resolution), and temporality classification.

- **Multi-Class NER:** Extracts MEDICATION (agents, formulations), DOSAGE (e.g. 500mg), FREQUENCY (e.g. twice daily with meals), DIAGNOSIS (e.g. Type 2 Diabetes Mellitus, Acute Pancreatitis), and LAB_OBSERVATION (e.g. HbA1c 8.2%, Lipase

210 U/L).

- **Negation Resolution (NegEx):** Evaluates a forward character window of 40 characters from trigger tokens ('denies', 'no history of', 'negative for', 'ruled out'). Extracted entities within the scope are assigned assertion_status = NEGATED.
- **Temporality Categorization:** Discriminates between HISTORICAL diagnoses ('history of', 'resolved', 'prior') and ACTIVE conditions ('current', 'active', 'taking').

5. Week 1 Verification Suite (tests/test_week1_pipeline.py)

Test Identifier	Scope & Assertions	Observed Output	Result
test_01_minio_connectivity	MinIO S3 connection, bucket creation, SHA-256 hash length == 64, payload round-trip.	`URI: ehr-raw-inputs/test/test-pipe-01.json, SHA256 verified`	**PASS (ok)**
test_02_ingest_speech_to_text	Ingest audio transcript, verify doc ID, status == INGESTED, 6 diarized fragments.	`doc.id == STT-2026-081234, len(fragments) == 6`	**PASS (ok)**
test_03_ingest_ocr_prescription	Ingest scanned prescription, verify document_class, bounding_box persistence.	`doc.id == OCR-2026-081234, bounding_box verified`	**PASS (ok)**
test_04_ingest_historical_doc	Verify quarantine invariant: status == PENDING_IDENTITY_RESOLUTION, patient_id == None.	`Quarantined doc with extracted Rohan Sharma hints`	**PASS (ok)**
test_05_ingest_hl7_api_json	Unpack FHIR/HL7 collection bundle into discrete clinical observation fragments.	`doc.id == HL7API-2026-081234, 6 fragments created`	**PASS (ok)**
test_06_nlp_extraction	Execute NER extractor over ingested narrative, verify MEDICATION and DIAGNOSIS extraction.	`entity_count > 0, MEDICATION and DIAGNOSIS extracted`	**PASS (ok)**